A verified 2-D baseline for the CAMR Pelanti-Shyue CO2 pipe-break

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Status: clean, symmetric, long-time 2-D evolutions established for single-phase and genuine two-phase regimes. Flash (nucleation) physics is deliberately **held out** for the next stage. This note documents the configuration, the incremental verification ladder, and the one remaining ingredient.

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1. Motivation

The target application is a compressible two-phase CO_2 release (a "pipe-break" blowdown) solved with the six-equation Pelanti–Shyue (PS) model on the CAMR/AMReX adaptive-mesh framework, using a Berger–LeVeque **wave-propagation** (`ps_flux=wp`) face solver with a Peng–Robinson real-fluid EOS. Earlier attempts to run the full flashing configuration (dense liquid vented into gas across the saturation dome) developed rapidly growing noise/asymmetry and eventually failed. This note isolates *which* physics is responsible and establishes a trustworthy baseline by turning on one ingredient at a time and verifying that each preserves a clean, mirror-symmetric, long-time solution before the next is added.

The governing invariant we enforce throughout: **every per-cell operator must be a continuous function of state**, so that round-off-level "chatter" stays at round-off and cannot be amplified into asymmetry or nonphysical states. This is essential for the downstream production cases, which will contain genuine noise.

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2. Configuration

A 2 m × 1 m rectangular domain (256×128 base cells; AMR refinement ratio 2). The x-lo face is the **rupture plane**, realised entirely in `bcnormal` (legacy path, `CAMR.ps_bc_use_nscbc=0`):

segment	boundary condition
x-lo break gap, $\lvert y-0.5\rvert \le 0.05$	**Dirichlet reservoir inlet** (soft-orifice, smoothstep lip taper over 0.02 m)
x-lo elsewhere	**reflecting slip wall** (normal momentum reflected)
x-hi, y-lo, y-hi	**linear-acoustic characteristic non-reflecting outflow** (far-field `p_amb`)

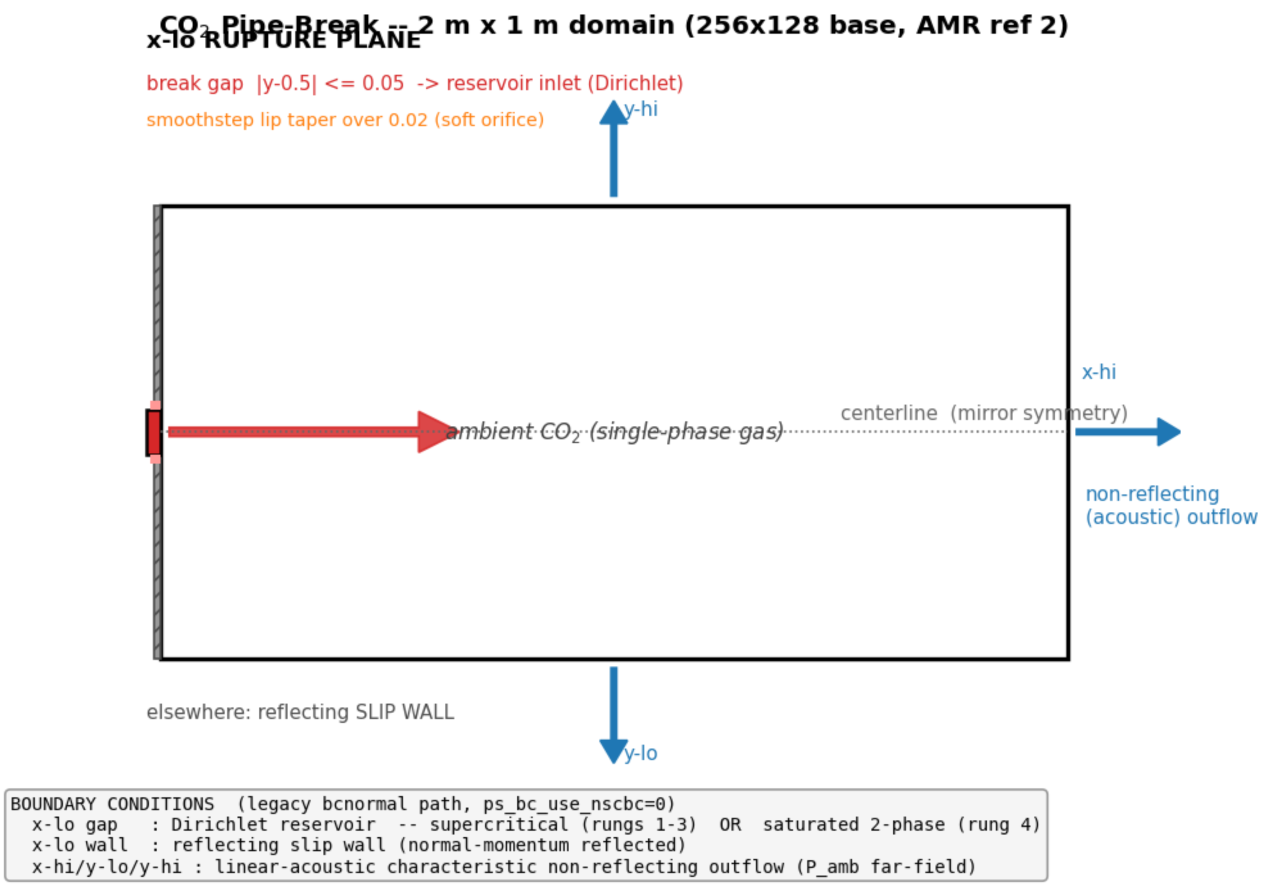

Figure 1: Configuration schematic

The setup is symmetric about the centerline `y = 0.5` (which lies exactly on a cell face for `n_cell_y = 128`), so **exact mirror symmetry is the correctness diagnostic**: any departure of the conserved fields from `f(y) = f(1−y)` above round-off signals an asymmetric numerical operator.

Numerics common to all cases: `ps_flux=wp` (wave-propagation, 2nd order via limited BL correction fluxes), `ps_recon=0` (piecewise-constant reconstruction — required for the reservoir-ghost injection), single-step unsplit advance (`do_mol=0`), CFL 0.3, continuous positivity floors on `P` and `T`, mechanical pressure relaxation on.

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3. Verification ladder

Each rung turns on one additional ingredient and is validated for symmetry (max|ρ(y)−ρ(1−y)|), boundedness (min P, min/max ρ, min T), and absence of NaN over a **long** time integration. All runs are round-off symmetric.

rung	ingredient added	case	result
1	single-phase supercritical jet, single level	`inputs.scjet` (T=400 K, 2:1)	520 steps / 3.0 ms; asym ρ ≤ 5×10^{-10}; quasi-steady
2	AMR (1 then 2 refinement levels)	`inputs.scjet`, `max_level=1,2`	symmetric with refinement active; C-F reflux/regrid clean
3	strong under-expansion (to 16:1)	`inputs.scjet`, T=600 K, `p_res` swept	symmetric barrel-shock / Mach-disk; 8:1 durable to 270 steps
4	**genuine two-phase** (mass transfer)	`inputs.satjet` (saturated blowdown)	270 steps / 2.2 ms; asym ρ ~4×10^{-11} (bounded); ~90 two-phase cells

Key points:

- **Rungs 1–3 (single phase).** To keep the fluid single-phase while raising the vent strength, the temperature is held above the critical temperature (400 K, then 600 K > T_crit = 304 K) so that an expansion cannot cross the saturation dome. The wave-propagation scheme resolves a clean barrel-shock/Mach-disk structure at 8:1–16:1 under-expansion while remaining exactly mirror-symmetric, confirming the multi-dimensional hydrodynamics, transverse coupling, boundary injection, and AMR coarse-fine machinery are all symmetry-clean.
- **Rung 4 (two-phase).** A new **saturated two-phase inlet** (`prob.res_alpha1`) injects an equilibrium liquid+vapor mixture (`EOS::co2_sat_LV` at `T_res`), so genuine two-phase cells enter the domain from the boundary and exercise the mass transfer and mechanical relaxation on real coexistence — **without** flash nucleation. The result is a clean, symmetric, long-time two-phase blowdown (below). This establishes that mass transfer + relaxation are well-behaved on two-phase cells; they are not the source of the earlier failures.

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4. What is held out for the next step

Flash (nucleation) is deliberately disabled (`ps_flash_tau = 0`) in every rung above. The reason is the central diagnostic result of this work:

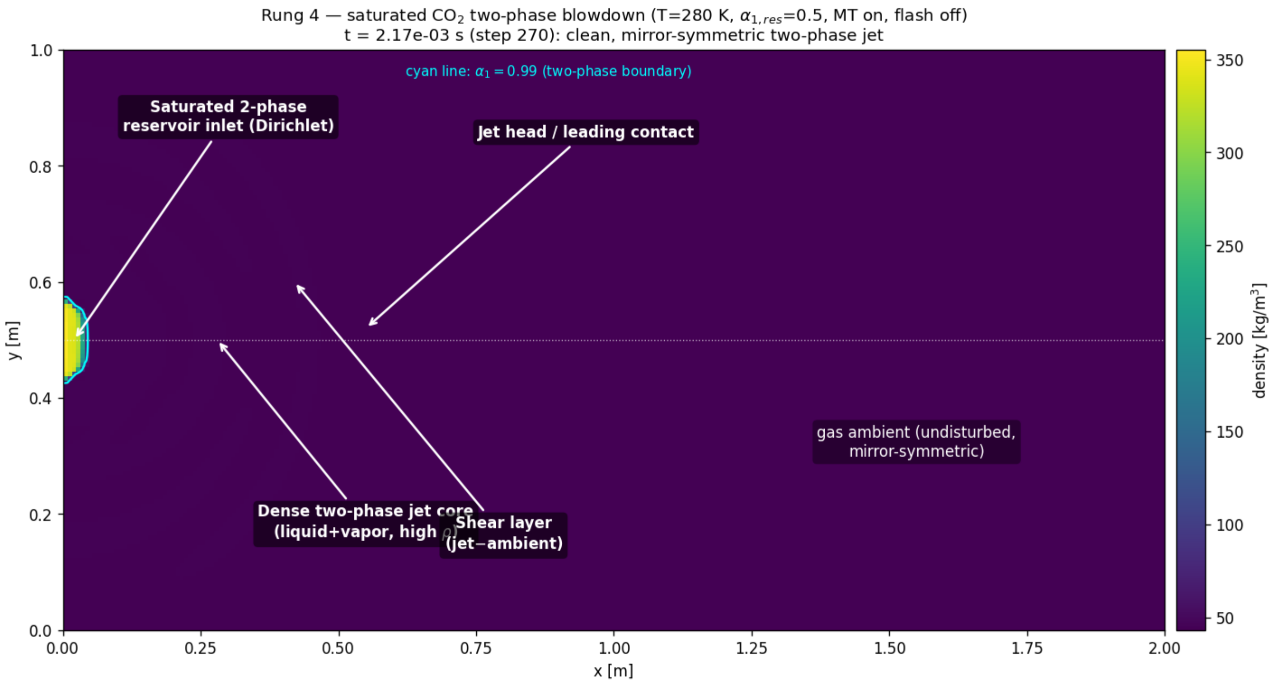

Figure 2: Annotated final two-phase solution

> In an Eulerian sharp-interface setting, **flash is the nucleation mechanism** that converts a metastable single-phase state into two phases. Without it, a metastable cell simply stays single-phase; with it, two-phase cells are *created* wherever the local state is metastable.

The flashing configuration places a metastable region exactly at the venting orifice lip — which is a strong hydrodynamic **amplifier** (an under-expanded jet lip). Flash there converts the lip to two-phase and the amplifier grows any perturbation, down to round-off, into an O(1) asymmetry within a few time steps. Making the flash source a *continuous* function of state (smooth activation windows in T, α, and the metastability margin; a fractional-flash limiter replacing the all-or-nothing EOS-validity rollback; finiteness-based gates that accept metastable $P \le 0$ branch states) reduced the injected seed by $\sim 10^3$ and delayed onset, but cannot by itself defeat a few-step-e-folding amplifier.

Therefore the next stage (**rung 5**) re-introduces flash behind a **centerline symmetry boundary condition** (half-domain), so nucleation physics can be developed and verified against the symmetric baseline without the lip instability masking correctness; and/or with explicit lip regularization. The deeper, fluid-agnostic route is a better-conditioned EOS evaluation at the saturation dome (an MLP surrogate) to remove the branch-locked validity discontinuity that the flash decisions inherit.

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5. Reproduction

```
# Rung 1–2: single-phase supercritical jet (add amr.max_level=1 or 2 for rung 2)
./CAMR2d.gnu.<...>.ex inputs.scjet

# Rung 3: strong under-expansion (deep supercritical)
./CAMR2d.gnu.<...>.ex inputs.scjet prob.T0=600 prob.T_res=600 prob.p_res=8.0e6
```

```
# Rung 4: saturated two-phase blowdown (mass transfer on, flash off)
./CAMR2d.gnu.<...>.ex inputs.satjet
```

Supporting figures in this directory: `pipebreak_schematic.png` (configuration), `scjet_montage.png` / `scjet_r8_structure.png` (single-phase rungs), `satjet_montage.png` / `satjet_annotated.png` (two-phase rung).